MILITARY SPECIFICATION

COMPRESSOR; PNEUMATIC, 3,000 PSI -- 2 SCFM

AERO 2A (*), AERO 2A-1 (*), AERO 2B (*), and AERO 2B-1 (*)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design and all performance requirements for an electrically-operated, self-contained, air compressor for use in aircraft armament pneumatic systems.

1.2 Classification - Pneumatic compressors shall be of the following Aero types as specified (see 6.2).

Aero 2A (*) - DC driven three-stage

Aero 2A-1 (*) - DC driven, four-stage

Aero 2B (*) - AC driven, three-stage

Aero 2B-1 (*) - AC driven, four stage

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)

FSC 4310

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MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specifications for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-P-5518	Pneumatic System; Design, Installation and Tests in Aircraft
MIL-C-5541	Chemical Films for Aluminum and Alluminum Alloys
MIL-L-6085	Lubricating Oil, Aircraft Instruments, Low Volatility
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-8879	Screw Threads, Controlled Radius Root With Increased Minor Dia[illegible]r, Specification for
MIL-M-7793	Meter, Time Totalizing, 380 to 420 cycle, General Specification for
MIL-M-7969	Motors, Alternating Current, 400 Cycle, 115/200 Volt System, Aircraft, Class A and Class B, General Specification for
MIL-M-8609	Motors, Direct Current, 28 Volt System, Aircraft, General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-N-18307	Nomenclature and Nameplate for Aeronautical, Electronic and Associated Equipment
MIL-P-23408	Plating: Tin-Cadmium(Electrodeposited)

STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards: Order of Precedence for the Selection of

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MIL-STD-794 Parts and Equipment, Procedure for Packing and Packing of
MIL-STD-704 Electric Power, Aircraft, Characteristics and Utilization of

MIL-STD-756 Reliability of Weapons Systems, Procedure for Prediction; and Reporting Prediction of

MIL-STD-781 Reliability Tests, Exponential Distribution

MS3102 Connector, Receptacle, Electric, Box Mounting

MS25245 Connections - Input Power Airborne Electronic Equipment; 28 - Volt DC, 115-Volt Single Phase AC, or 115/200-Volt Three Phase AC, or combinations

MS33586 Metals, Definition of Dissimilar

MS33657 Fitting End, Standard Dimensions for Bulkhead Flared Tube Connection

MS33656 Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

2.2 Availability of Documents

*2.2.1 When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

*3.1 Preproduction - The compressor furnished under this specification shall be a product, samples of which have been tested and have passed the preproduction test specified herein.

*3.2 Selection of Government Documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial Parts - Commercial parts having suitable properties shall be used, when, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, cotter pins, having suitable properties, shall be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and on the contractor's drawings.

3.2.2 Standard Parts - With the exception of 3.2.1 MS or AN Standard

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parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Critical Materials - Non-critical materials shall be used where practicable. Where the use of a critical material is essential to meet specifications requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Metals - Metals shall be of the corrosion resistant type or suitably treated to resist corrosion caused by fuels, salt spray, lubricants, or atmospheric conditions likely to be met in storage or normal service.

3.3.2.1 Dissimilar Metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

*3.3.2.2 Magnesium Alloy Parts - Magnesium alloy parts shall be treated in accordance with Specification MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

*3.3.2.3 Aluminum Alloy Parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

*3.3.2.4 Iron and Steel Parts - Iron and steel parts which are in hermetically sealed cases shall be cadmium plated in accordance with Specification QQ-P-416. Parts which are not in sealed cases shall be chromium, nickel or zinc plated in accordance with Specification QQ-C-320 or QQ-N-290 or QQ-Z-325 respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated in accordance with specification MIL-P-23408.

3.3.3 Protective Treatment - When materials are used in the construction of the equipment that are subject to deterioration when exposed to climate and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance of this specification. Any protective coating that will crack, chip, or scale with age or extremes of climate and environmental conditions shall not be used.

*3.3.4 Fungus - Proof Materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they

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shall be treated with a fungicidal agent acceptable to the procuring activity.

*3.3.5 Fumes and Vapors - Materials used in the construction of the compressor shall not produce corrosive, deleterious or toxic fumes, or vapors under the conditions specified herein.

*3.4 Design and Construction - In the design and construction of this equipment, fulfillment of performance objectives shall be the prime consideration. All components within the scope of their specific application as a part of the equipment shall conform to Specification MIL-P-5518 except as otherwise specified herein.

*3.4.1 Provisions for Maintenance and Field Testing - The equipment shall provide for ease of maintenance to the greatest extent practicable. Provisions for field testing shall be as specified in Specification MIL-E-5400.

3.4.2 Screw Threads - Screw threads 0.060 inch in diameter or larger shall be in accordance with MIL-S- 8879.

*3.4.3 Time Meter- Provisions shall be made for a time totalizing meter in accordance with Specification MIL-M-7793 to record the total operating time of the equipment. The meter shall be of the digital type and the maximum read-out shall exceed the minimum total operating life specified herein for the equipment.

3.4.4 Electrical Requirements

3.4.4.1 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) Aero 2A (*) and Aero 2A-1 (*)
DC Power 28V, Category "B", 70 Amps

(2) Aero 2B (*) and Aero 2B-1 (*)
DC Power 28V, Category "B", 3 Amps
AC Power (Three Phase) 115/200V, Category "B", 3640 VA

3.4.4.2 Input Power Connection - Input power connection shall be designed in accordance with Drawing MS25245.

3.4.4.3 Protection -

3.4.4.3.1 Voltage Protection - The equipment shall not be damaged by voltages either above or below the voltages specified herein and shall resume normal operation automatically when voltage returns within designed limits.

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3.4.4.3.2 Overload Protection - All parts and circuits of the equipment which are likely to carry an overload due to failures and maladjustments shall be protected with suitable resetable protective devices.

3.4.4.4 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet electromagnetic interference control requirements and provided that the assembly of the cable and its connectors may be accomplished easily.

3.4.4.5 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. However, cables and those portions of the connections that are to be attached to the cables shall not be supplied as part of the equipment. Wiring between electrical components shall be coded in accordance with specification MIL-W-5088.

*3.4.5 Oil Tank - A tank shall be provided to furnish lubricating oil to the equipment. The capacity of the tank shall be sufficient to furnish lubricating oil for at least 30 hours of equipment operating time. The oil tank shall be removable without dismounting the compressor from its installed position. A sight gauge or other means shall be provided for visual indication of the oil level. The manufacturer as part of its maintenance procedures shall state the maximum acceptable oil consumption.

*3.4.6 Lubrication - The equipment shall be lubricated with oil conforming to Specification MIL-L-6085.

* 3.4.7 Moisture Separator - A moisture separator shall be provided to remove at least 96% of the moisture and oil vapors from the discharge air. The separator shall be equipped with a heater which shall have been energized when the ambient temperature reaches 0°C. The spread between the energized and the de-energized position shall not be less than 8.3°C and not more than 20°C.

3.4.7.1 Condensate Dump Valve - A condensate dump valve shall be provided to dump the accumulated moisture collected in the moisture separator. The valve shall be integral with the moisture separator or closely adjacent to minimize the possibility of freezing during low temperature operation. The valve shall be activated as necessary to preclude discharge of accumulated moisture and oil into the pneumatic system.

*3.4.8 Pressure Relief Valve - A pressure relief valve shall be provided which shall by-pass full air flow of the compressor to the atmosphere when a maximum gage pressure of 3750 PSI is reached. The pressure relief valve shall reset at a minimum pressure of 3200 PSI gage.

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*3.4.9 Pressure Control Switch - A pressure control switch shall be provided which shall cut out the compressor when the outlet pressure reaches 3100 ±100 PSI gage and cut in the compressor when the outlet pressure drops to 2700 ±100 PSI gage.

*3.4.10 Back Pressure Valve - For efficient operation of the moisture separator, the compressor shall be equipped with a back pressure valve with an operating pressure of 2550 ±150 PSI full flow pressure.

3.4.11 Detail Requirements

3.4.11.1 Aero 2A(*) and Aero 2A-1(*)

3.4.11.1.1 Form Factor - The envelope dimensions of the equipment shall not exceed those shown in Figure 1.

3.4.11.1.2 Weight - The dry weight of the equipment, excluding external wiring, shall not exceed 40 pounds.

3.4.11.1.3 Mounting - The mounting dimensions of the equipment shall be in accordance with Figure 1.

*3.4.11.1.4 Electrical Connection - Connections to external circuits shall be provided by an MS3102R-20-24P connector (J101) located as shown in Figure 1 and connected as follows:

Pin Designation	Function
A	Control
B	28VDC
C	Spare
D	Ground

*3.4.11.1.5 Pneumatic Connections - The equipment shall have MS33656 and MS33657 connections located as shown in Figure 1. Connections shall withstand tubing installation torque without movement or leakage.

3.4.11.1.5.1 Air Outlet - The air outlet pneumatic connection shall be for 1/4 inch tubing.

3.4.11.1.5.2 Condensate Outlet - The condensate outlet pneumatic connection shall be for 1/8 inch tubing.

3.4.11.1.6 Motor - The equipment shall be powered by a continuous duty electric motor utilizing 28V DC and conforming to Specification MIL-M-8609. The motor shall be deemed adequate if it operates in an acceptable and satisfactory manner during testing of the complete equipment.

3.4.11.2 Aero 2B(*) and Aero 2B-1(*)

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3.4.11.2.1 Form Factor - The envelope dimensions of the equipment shall not exceed those shown in Figure 2.

3.4.11.2.2 Weight - The dry weight of the equipment, excluding external wiring, shall not exceed 33 pounds.

3.4.11.2.3 Mounting - The mounting dimensions and air intake of the equipment shall be in accordance with Figure 2.

*3.4.11.2.4 Electrical Connection - Connections to external circuits shall be provided by two connectors located as shown in Figure 2 and connected as follows:

Reference Designation	Receptacle Type	Pin Function
J101-1-A	MS3102R-10SL-3P	28 V DC
J101-2-B		Ground
J101-3-C		Control
J102-1-A	MS3102R-18-4P	Phase 1 115 V AC
J102-2-B		Phase 2 115 V AC
J102-3-C		Phase 3 115 V AC
J102-4-D		Ground

*3.4.11.2.5 Pneumatic Connections - The equipment shall have MS33656 and MS33657 connections located as shown in Figure 2.

3.4.11.2.5.1 Air Outlet - The air outlet pneumatic connections shall be for 1/4 inch tubing.

3.4.11.2.5.2 Condensate Outlet - The condensate outlet pneumatic connection shall be for 1/8 inch tubing.

3.4.11.2.6 Motor - The equipment shall be powered by a continuous duty electric motor utilizing 400 Hz 3 phase 115/200 V AC and conforming to Specification MIL-M-7969. The motor shall be deemed adequate if it operates in an acceptable and satisfactory manner during testing of the complete equipment.

*3.5 Performance - The compressor shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

*3.6 Reliability - The equipment shall be capable of the total operating life and mean-time-between-failures specified herein. Equipment reliability shall be analyzed, calculated, and reported in accordance with MIL-STD-756. The equipment shall have a mean-time-between-failures of 250 hours when tested and accepted under the provisions of paragraph 4.4.3 and paragraph 4.6.16.

A "failure" is defined as the compressor unit:

(1) not operable

(2) operable but not in conformance with air capacity requirements.

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(3) excessive oil consumption
(4) excessive power demand
(5) structure failures which may become major and lead to the above failures

3.7 Nameplates - A nameplate conforming to Specification MIL-N-18307 shall be securely attached to the exterior of the compressor and shall be marked in accordance with the requirements of Military Standards MIL-STD-130.

3.8 Finish - The equipment shall be finished in durable glossy black color.

3.9 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Parts joined together as by welding or riveting shall be interchangeable as assemblies. Changes in manufacturer's part numbers shall be governed by the item identification and part numbering requirements of Specification MIL-D-1000.

3.10 Workmanship - The equipment, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, plumbing, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

3.10.1 Dimensions - Dimensions and tolerances not specified shall be close as is consistent with best shop practice. When dimensions and tolerances affect the interchangeability, operation, or performance of the equipment, the dimensions and tolerances shall be held or limited accordingly.

3.10.2 Fabrication - Machining, drilling, and forming shall be done with the use of accurate jigs, templates, or gages.

3.10.3 Screw-assemblies - Assembly screws shall be tight. The word "tight" means that the screws cannot be tightened appreciably further without damage or injury to the screw threads.

3.10.4 Riveting - Riveting operations shall be performed carefully so that the rivets are tight. Rivet heads shall be full size and concentric with the body. Dimensions of the rivets shall be such that the holes will be filled. Loose, cracked, or badly formed rivets shall be replaced before acceptance of the equipment. Excess metal shall be removed from protruding counter-sunk rivet heads as a customary operation. This does not cover eyelets or hollow tubular rivets which shall not be used.

3.10.5 Gears - Gear assemblies shall be aligned and meshed properly, and shall be operable without interference, tight spots, loose spots, or other irregularities. When required for accurate adjustment, gear assemblies shall be free from detrimental backlash.

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3.10.6 Wire Stripping - Wire stripping shall be done without nicking or otherwise damaging the conductor.

3.10.7 Cleanup - The equipment shall be cleaned thoroughly of loose parts, spattered or excess solder, metal chips, and other foreign material after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractors' Responsibility - The supplier is responsible for the performance of all requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Classification of Tests - The Inspection and Testing of the compressor shall be classified as follows:

4.2.1 Preproduction Tests - Preproduction tests are those tests performed after award of contract on two compressors of each type under procurement to determine that the item meets the requirement of this specification. The compressors shall be representative in design, construction, quality, materials, workmanship, method of manufacture, performance, and configuration of the units which will be produced on the manufacturer's production line. Deviation from the standards of the preproduction sample shall be subject to the approval of the procurement activity. Evidence of unauthorized change shall constitute cause for rejection.

4.2.2 Quality Conformance Tests-Conformance tests are those tests performed on compressors manufactured and submitted for acceptance under contract.

4.2.3 Production Tests - Production tests shall be performed on two units of each type procured. They shall be selected at random from the first fifteen units produced on the manufacturer's production line. The units are to be subjected to any or all tests of this specification as determined by the procuring activity.

4.3 Preproduction Tests - The preproduction tests of the compressor shall consist of all of the tests of this specification specified under paragraph 4.4.

4.3.1 Preproduction Test Sample Instructions - Preproduction test samples shall consist of two compressors of each type manufactured in accordance with this specification. The samples submitted shall not have been previously tested except for the acceptance tests. Samples shall be fitted with external connections for (1) bypassing the function of the pressure switch, and (2) a pneumatic connection prior to the back pressure valve. These units shall be

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forwarded to the testing laboratory designated in the contract or order. The contractor shall submit with his units, two copies of the maintenance procedures for each type of unit procured. These procedures shall be approved, in advance, by the procuring activity.

4.3.2 Preproduction Test Sample Identification - The test samples shall be plainly identified by securely attached, durable tags marked with the following information:

Sample for Preproduction Testing
Pneumatic Compressor, Aero 2--
Submitted by (manufacturer) on (date) for preproduction testing in accordance with Specification MIL-I-7135 under (Contract or Order Number)
Manufacturer's Part Number _____

4.4 Quality Conformance Test-The conformance tests shall consist of the individual tests, sampling plan tests and Reliability Assurance of this Specification. The contractor shall furnish all samples and shall be responsible for accomplishing the conformance tests. Conformance tests shall be under the supervision of the Government Inspector. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual Tests - Individual tests shall be conducted at the contractor's plant on each equipment submitted for acceptance under the contract and shall consist of the following:

(1) Examination (Para. 4.6.1)
(2) Weight (Para. 4.6.2)
(3) Capacity (Para. 4.6.3)
(4) Operational Check (Para. 4.6.4)
(5) Pressure Drop (Para. 4.6.7)

4.4.2 Sampling Plan Tests - The test samples selected for sampling tests shall first have passed the individual tests. Test samples shall not be delivered on contract until they have been refurbished and resubmitted and passed all the individual tests.

4.4.2.1 Sampling Plan Sample Selection - Sampling Plan Samples shall be selected at random in accordance with the following schedule:

Quantity Offered For Acceptance	Quantity to be Selected for Testing
99 or less	1
100 to 299	3
300 to 499	4
500 or more	4 plus one for each additional 500 or fraction thereof

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4.4.2.1.1 Sampling Plan Tests - Each sample selected for Sampling Plan Testing shall be subjected to the following tests:

(1) High Temperature Operation (Para. 4.6.8.2)
(2) Low Temperature Operation (Para. 4.6.9.2)
(3) Altitude (Para. 4.6.10)
(4) Proof Pressure (Para 4.6.11)

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781 as a guide. The following temperature test structure and vibration shall be used in lieu of those as specified in test Level E.

a) One unit out of every ten tested at -18°C
b) One unit out of every ten tested at +38°C
c) The remaining units tested at standard conditions
d) No vibration shall be applied except that the units shall be hard mounted.

*4.4.3.1 Lot Size and Test - Each equipment produced shall be tested for 20 hours. In addition to the 20 hours a run-in period may be used at the option of the contractor. If used, the run-in must be the same for each equipment, in time and severity. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan V. MIL-STD-781. (Extend the line as necessary to accomodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command Attn: Armament Division. At any time that the current totals of test hours and test failures plotted on Test Plan V curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

*4.4.4 Production Test Instructions - Two compressors shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded to the laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information:

Pneumatic Compressor, Aero 2
Submitted by (manufacturer) on (date) for production testing; in accordance with Specification MIL-C-7135, under (Contractor Order Number)
Manufacturer's Part Number _____

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4.5 Test Conditions

*4.5.1 Standard Conditions - The following room conditions shall be used as a basis to establish normal performance characteristics and for making laboratory bench tests on the equipment.

Temperature	25° ± 5°C
Relative Humidity	90% or less
Barometric Pressure	28 to 32 inches of mercury
Vibration	None
Input Power Voltage	115 ± 1.0 V AC 3 phase or 27.5 ± 0.5 V DC

4.5.2 Attitude - Unless otherwise specified the compressor shall be tested in its normal operating position.

4.5.3 Test Details - The equipment duty cycle shall consist of 3 1/2 hours time-on and 1/2 hour time off. During the time-on the equipment shall deliver 2 SCFM at 3,000 ±75 PSIG discharge pressure with inlet at standard conditions.

*4.5.4 Tolerances - The maximum allowable tolerances on the test conditions shall be as follows:

a) Temperature: Plus or minus 2°C (3.6°F). (Exclusive of accuracy of instruments.)
b) Altitude: Plus or minus 5 percent (in feet).
c) Relative humidity: Plus 5 percent, minus 0 percent (of R.H. value)

*4.5.5 Measurements - All measurements shall be made with instruments the accuracy of which conforms to acceptable laboratory standards, and which are appropriate for measurement of environmental condition concerned. If tests are conducted at the contractor's plant, the accuracy of the instruments and test equipment shall be verified periodically by the contractor to the satisfaction of the procuring activity.

4.6 Test Methods

4.6.1 Examination of Product - Each compressor shall be examined externally to determine conformance with the requirements of this specification not covered by tests.

4.6.2 Weight - The dry weight of the compressors excluding internal wiring, shall not exceed the following:

a.	Aero 2A and Aero 2A-1	40 pounds
b.	Aero 2B and Aero 2B-1	33 pounds

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4.63 Capacity

*4.6.3.1 Sea Level Altitude - At room temperature and sea level ambient pressure altitude (29.92 inches of mercury), the equipment shall charge a cylinder having a capacity of 100 cubic inches to a pressure of 3000 PSIG in a maximum time of 7 minutes.

*4.6.3.2 40,000 Feet Altitude - At an ambient pressure of 5.8 inches of mercury and an ambient temperature of -18°C (0°F) the equipment shall charge a cylinder having a capacity of 100 cubic inches to a pressure of 3,000 PSIG in a maximum time of 42 minutes.

4.6.4 Operational Check

4.6.4.1 Aero 2A(*) and Aero 2A-1(*) - The equipment shall be connected to a variable voltage source and checked for starting and operation at 21 volts and 29 volts with a discharge pressure of 2700 ± 100 PSIG.

4.6.4.2 Aero 2B (*) and Aero 2B-1 (*) - The equipment shall be connected to a variable frequency voltage source and checked for starting and operation at 380 Hz, 106 volts; at 400 Hz, 106 volts; at 420 Hz, 106 volts; and at 380 Hz, 120 volts with a discharge pressure of 2700 ±100 PSIG.

4.6.5 Insulation Resistance - A constant potential of 500 volts direct current shall be applied between each electrical circuit and all other electrical circuits and between each circuit and the structural ground for a duration of not more than one minute. A measured insulation resistance of less than one megohm as shown by a megger or other suitable method shall be a failure to pass this test. See note in paragraph 4.6.6.

4.6.6 Dielectric Strength - An electro-motive force of at least 1000 volts, root mean square, alternating current, shall be applied between each electrical circuit and all other electrical circuits and between each circuit and the structural ground for duration of one minute. The frequency of the applied voltage shall be in the range of 50 to 100 Hz. Breakdown, puncturing of insulation or arcing as shown by a suitable indicating device shall be a failure to pass this test. Note: If the unit contains components such as RFI filters and solid state devices which will be damaged by paragraphs 4.6.6 and 4.6.5 tests such components shall be identified in supplied maintenance procedures.

4.6.7 Pressure Drop Test - With the compressor outlet connected directly to a pressure gauge the unit is started and allowed to run until it is automatically shut off by the pressure regulation system at a pressure of 3100 ±100 PSIG. The pressure drop shall not exceed 40 PSI per minute averaged over the time from cut out to a pressure 400 PSIG below the cut out pressure.

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4.6.8 High Temperature Test

*4.6.8.1 High Temperature Exposure - The compressor shall be tested in accordance with Procedure II of Specification MIL-E-5272 except the temperature shall be maintained at +85°C (185°F). The equipment shall not be operated at the high temperature but shall operate satisfactorily at standard conditions with rated voltage after completion of the High Temperature Tests.

4.6.8.2 High Temperature Operation - The compressor shall be properly connected except that no power shall be applied. The compressor shall then be subjected to an ambient temperature of 74°C (165°F) for a period of 4 hours. During the last 10 minutes of the 4 hour period power shall be applied and the compressor shall operate satisfactorily at rated voltage and delivery pressure. (This test may be combined with the High Temperature Exposure Test at the discretion of the testing agency). Note: Compressor capacity allowable: 100 IN^3 to 3000 PSIG in 8.5 minutes maximum.

4.6.9 Low Temperature Test

*4.6.9.1 Low Temperature Exposure - The compressor shall be tested in accordance with Procedure II of Specification MIL-E-5272 except the -62°C (-80°F) temperature shall be maintained for 48 hours. The equipment shall not be operated at the low temperature but shall operate satisfactorily at standard conditions with rated voltage after completion of the Low Temperature Test.

*4.6.9.2 Low Temperature Operation - The compressor shall be properly connected except that no power shall be applied. The compressor shall then be subjected to an ambient temperature of -54°C (-65°F) for a period of 4 hours. During the last 10 minutes of the 4 hour period power shall be applied and the compressor shall operate satisfactorily at rated voltage and delivery pressure. (This test may be combined with the Low Temperature Exposure Test at the discretion of the testing agency). Note: Capacity - 100 IN^3 to 3000 PSIG in 4.8 minutes maximum. The moisture separator heater (if fitted) may be energized at 3 1/2 hours of the 4 hour period.

*4.6.10 Altitude - The compressor shall be placed in a chamber and properly connected, except that no power shall be applied. The compressor shall then be subjected to an ambient temperature of -18°C (0°F) and then the internal chamber pressure decreased to 3.44 inches of mercury (50,000 ft). After an hour period the internal pressure of the chamber shall be increased to 5.8 inches of mercury (40,000 ft.). After the chamber pressure has stabilized the compressor shall charge a cylinder having a capacity of 100 cubic inches to a pressure of 3,000 PSIG in a maximum time of 42 minutes. (This test may be combined with the Low Temperature Testing at the discretion of the testing agency).

*4.6.11 Proof Pressure - The equipment shall be proof pressure tested in accordance with Specification MIL-P-5518. The proof pressure test shall be conducted with the equipment running, the pressure switch by passed and then by manipulating a valve in the discharge line to unseat and reseat the relief valve. The relief valve pressures shall be recorded to determine compliance with these requirements.

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4.6.12 Electromagnetic Interference Test - The electrical system of the compressor shall be tested for conducted and radiated electromagnetic interference (EMI) in accordance with Specification MIL-I-6181 over a frequency range of 0.35 mHz to 25 mHz for conducted interference, and 0.15 mHz to 400 mHz for radiated interference.

*4.6.13 Temperature Shock - The compressor shall be tested in accordance with Procedure I of Specification MIL-E-5272. At the conclusion of the third cycle, and after return to a standard condition the equipment shall operate satisfactorily at rated voltage and delivery pressure.

*4.6.14 Sand and Dust - The compressor shall be tested in accordance with Procedure I of Specification MIL-E-5272. At the conclusion of the test and after return to standard conditions the equipment shall operate satisfactorily at rated voltage and delivery pressure.

*4.6.15 Vibration - The compressor shall be subjected to random vibration applied to each of the three mutually perpendicular axes. When using manual equalization, the random vibration shall be applied at 20% of test values for sufficient time to adjust the filters. The equalized frequency spectrum shall be continuous and uniform within $\pm$ 3 db over the frequency range of 20 to 2,000 Hz as determined by an analysis using a band width of 25 Hz or less (and a sufficient sweep speed and appropriate averaging resulting in an accuracy equivalent to 120 degrees of freedom). Random vibration at .05g Hz over the 20 to 2,000 Hz bandwidth (10g rms overall) having Gaussian distribution and a peak to rms G ratio of 3 to 1 shall be applied to the equipment for a period of 80 minutes in each axis. There shall be no damage and the equipment shall not fail to function as a result of this test. The compressor shall be operated during vibration with a discharge pressure of 2900 $\pm$100 PSIG.

4.6.16 Life - The compressor shall be properly connected and operated for a period of 1000 hours. Of this total period 100 hours shall be conducted with an ambient temperature of -18°C (0°F), 100 hours with an ambient temperature of +38°C (+100°F), and the remaining 800 hours at room ambient. The compressor shall be cycled during the entire 1000 hours. The duty cycle shall be 3 1/2 hours operating and 1/2 hour not operating. Reliability conformance to paragraph 3.6 will be demonstrated as part of this test. At the completion of the 1000 hours endurance the compressor shall operate satisfactorily at rated voltage and delivery pressure. Deterioration of capacity of the compressor shall not exceed 10% of the rated 2 scfm and shall be linear over the 1000 hours endurance test. The contractor shall submit with his units, maintenance procedures to be followed during testing. Said procedures shall be approved by the procuring activity.

4.6.17 Humidity - The compressor shall be tested in accordance with Procedure I of Specification MIL-E-5272. At the conclusion of the test there shall be no evidence of corrosion or rusting of the internal or external parts of the equipment that would adversely affect subsequent operation. Within one hour after completion of the test the compressor shall satisfactorily demonstrate operation at rated voltage and delivery pressure.

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4.6.18 Fungus Resistance - The compressor shall be tested in accordance with Procedure I of Specification MIL-E-5272 except that the time period shall be 14 days. At the completion of the test there shall be no deterioration that would adversely affect subsequent operation. Any evidence of growth of fungi shall be cause for rejection.

4.6.19 Salt Spray - The compressor shall be tested in accordance with Procedure I of Specification MIL-E-5272 except the test period shall be 50 hours. At the end of the test period the compressor shall operate satisfactorily at rated voltage and delivery pressure.

4.6.20 Explosion - The compressor shall be tested in accordance with Procedure III of Specification MIL-E-5272. Explosive vapor shall not be inducted into the inlet. Any explosion caused by the test item shall be cause for rejection.

4.6.21 Shock - The compressor shall be tested in accordance with Procedure V of Specification MIL-E-5272. (Note: equipment operational portion may be performed in conjunction with the Vibration Test.)

*4.6.22 Moisture Separator - At an airflow of 2 SCFM containing .04 pound of water per pound of air, and an oil flow of the maximum rate allowed by paragraph 3.4.5 (that is the rate which will reduce a full oil tank to the minimum safe level in 30 hours) the moisture separator shall remove 96% by weight of the moisture and oil from the discharge air. The test shall be conducted at 3000 PSIG and 100°F. The separator may be removed from the compressor for this test or conducted on individual samples of separators. The time cycle of the condensate dump valve shall be measured to insure compliance with 3.4.7.1.

4.7 Rejection and Retest

*4.7.1 Defective Units - All defective parts, assemblies, or equipments shall be rejected.

4.7.2 Lots - Whenever a sample unit fails to meet any requirement, the lot represented by the sample unit shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 General - Each compressor shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-STD-794. All Level A containers shall be lined with a case liner.

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6. NOTES

6.1 Use - The equipments covered by this specification are intended for use in aircraft armament pneumatic systems conforming to Specification MIL-P-5518 at an operating pressure of 3,000 PSIG regulated to lower pressures at the inlet of the operated equipments.

6.2 Ordering Data - Procurement documents shall specify the following:

(1) The quantity and Aero type equipment (see 1.2) to be procured in accordance with this specification.

(2) The level of shipment required. (see 5.1)

(3) Title, number and date of this specification.

6.3 Definitions - For the purpose of this specification the following definitions apply:

(1) Equipment - The word "equipment" is used to mean the Pneumatic Compressors, Aero 2A, Aero 2A-1, Aero 2B and Aero 2B-1 covered by this specification.

6.4 The parentheses (*), when used in the type designation, may be replaced by either a number, a letter, or a new designation furnished by the procuring activity upon application by the contractor for assignment of nomenclature. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.5 The asterisk preceding a paragraph number denotes a change in that paragraph from the "E" version of this specification.

6.6 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the precedenct specified herein shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications, shall be approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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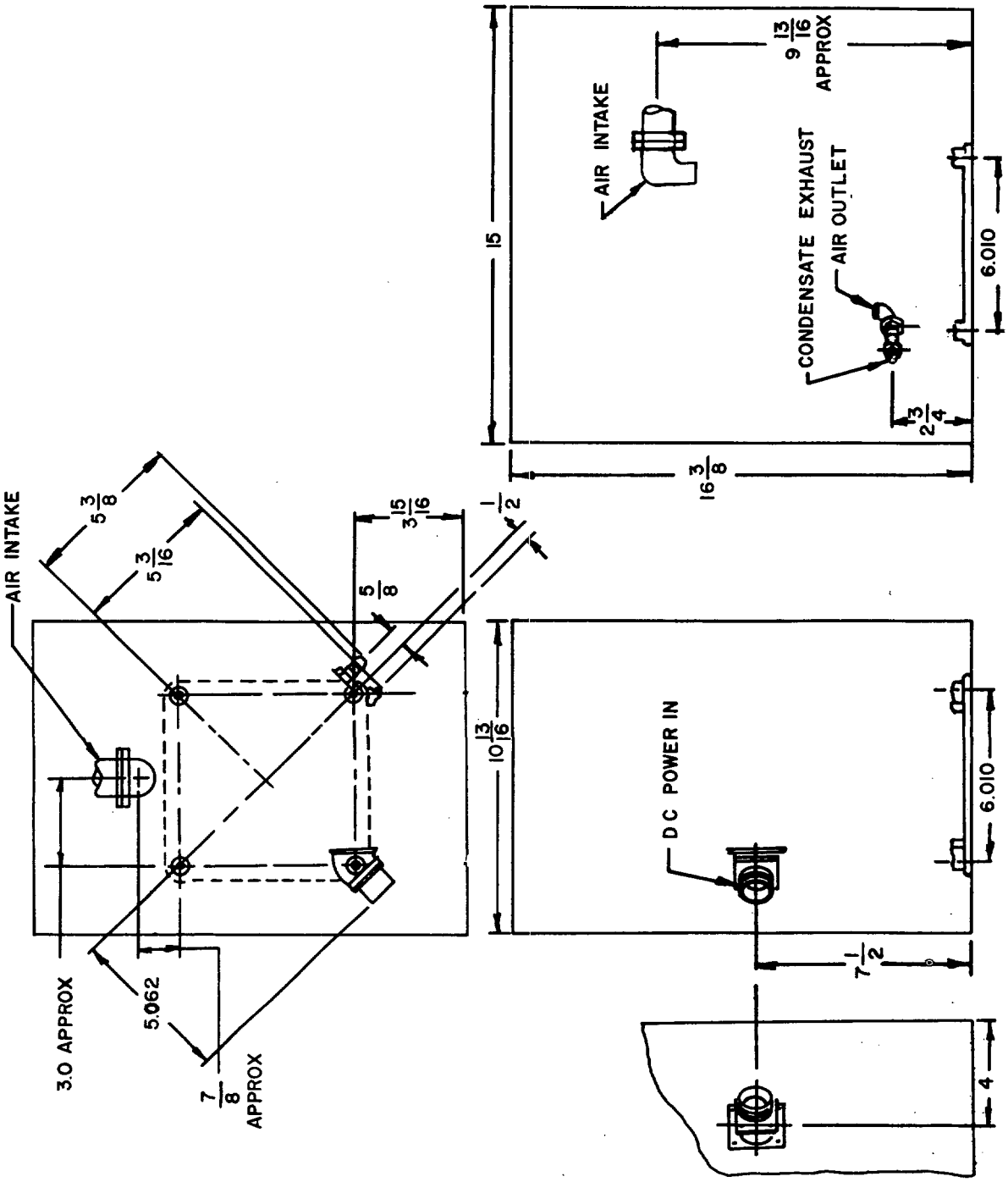


FIGURE 1

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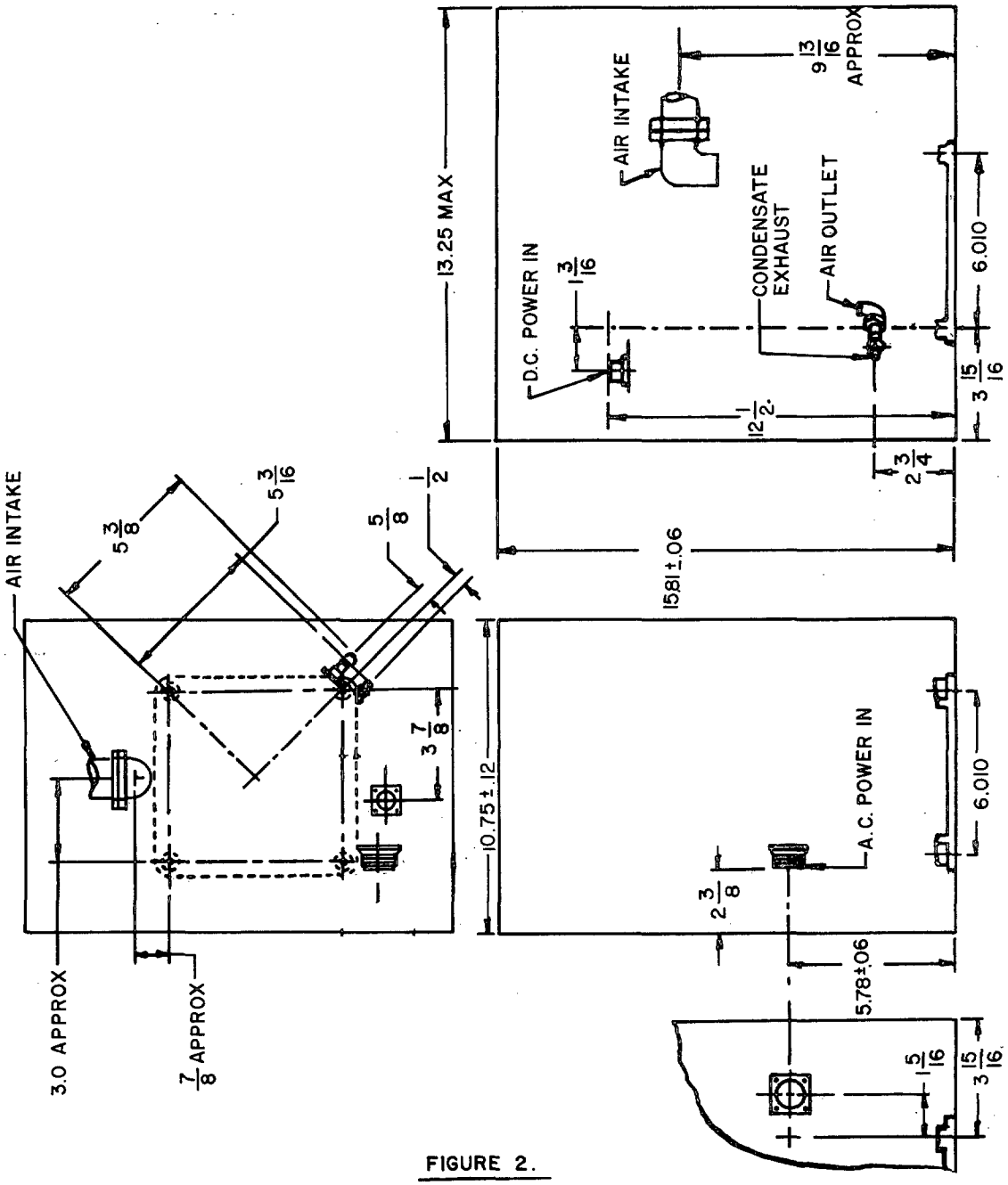


FIGURE 2.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-7135F(AS) COMPRESSOR: PNEUMATIC, 3,000 PSI--2 SCFM Aero 2A(*), Aero 2A-1(*), Aero

ORGANIZATION *(Of submitter)* 2B (*), and AERO 2B-1 (*)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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